

Quiz 1 Study Guide

The quiz will be in-class on Friday, February 6. It will be 15 minutes long. Some combination of the following problems will be on the quiz. You do not need to turn this study guide in; it is purely for your benefit. On the exam, please aim to answer the questions as precisely as you can (when grading your answers, I will be looking for mathematical definitions, not examples).

Problem 1. What does it mean for two events A and B to be **mutually exclusive**?

Problem 2. In order for a function $\mathbb{P}$ to be a **probability measure**, it must satisfy three axioms. State all three axioms.

Problem 3. Let A and B be events, and assume that $\mathbb{P}[A] > 0$. What is the **conditional probability of B given A** ?

Problem 4. What is a **k -tuple**?

Problem 5. Describe the difference between a **combination** and a **permutation**.

Problem 6. A **set** is an _____ collection of elements, all of which are _____.

Problem 7. What does it mean for two events to be **disjoint**?

Problem 8. What is the **sample space** of an experiment? What is an **event**?

Problem 9. What does it mean for a sequence of events $E_1, E_2, E_3, \dots$ to be **pairwise disjoint**?

Problem 10. Let A and B be events. Define the following terms

- the **complement** of A
- the **union** of A and B
- the **intersection** of A and B

Problem 11. What does it mean for two events A and B to have a **positive relationship**? A **negative relationship**?